

Syllabus for Introduction to Thermodynamics and Statistical Mechanics (PHY 404)

Instructor: Nathan Schine (nschine@umd.edu, PSC 2160)

TA: Mingyi Xu (mingyixu@umd.edu)

Room and Time: MWF 9:00-9:50, Microbiology 1207

Office hours: Fridays 1:00-2:00 pm, or by appointment, or drop-in any time.

Office hours will be held in my office PSC 2160, or with enough attendance, in PSC 2148, which is a conference room directly across from my office.

Textbook: "An Introduction to Thermal Physics" by Daniel V. Schroeder.

Additional reference texts: "Thermal Physics" by Kittel and Kroemer; "Fundamentals of Statistical Physics and Thermal Physics" by F. Reif

Course goals: The goal of this class is to introduce you to the basic laws of thermodynamics, the basic principles of statistical mechanics, and the statistical way of modelling the world.

Course Format: This is a lecture-based class for which your **in-person attendance is mandatory**. Significant sections of the class will go beyond the textbook, so taking detailed notes from the lectures will be essential. Class notes will not be posted on ELMS. If you do miss class, it is your responsibility to find out about assignments and topics covered in class. One class per week will typically (but not always) be converted into a problem solving session, focusing on solutions to problem set questions that have just been turned in.

Grades: Your grade will be based on problem sets (10%) of which one will not count, participation (5%), two in-class midterm exams (25% and 25%), and one final exam (35%). You are encouraged to work on the homework in groups but the final write-up submitted must be entirely your own. **Homework and exams have fundamentally different formats and purposes: homeworks help you learn over multiple hours, tests evaluate what you have learned in a short period of time.** So don't expect the homework to be very similar to the tests.

The conversion between numerical assessment grades and letter course grades is not preset and will take into account inevitable variation in the difficulty of the exams.

Exam policies: This policy is implemented not only to avoid cheating, but also to avoid the appearance of possible cheating. Exams will be in-class, closed-book, closed-notes, no calculator or electronic devices of any kind allowed. This means no tablet, cell-phone, or headphones/earbuds may be visible, much less used. If you leave the room your exam will be collected, and you will not be permitted to re-enter the room. Water is allowed, but be careful with this since you won't be able to use the bathroom. Food is not allowed other than cough-drops or similar.

Homework policies:

(a) Problem sets are due at 9 am each Monday. The full problem set should be submitted electronically ahead of time as a single pdf or image file, and then brought with you in person to class.

(b) Late submissions are **not** accepted without an extension. Extensions will be granted on a case-by-case basis and only if arranged more than a week ahead of time. As soon as you know you need an extension, please request it. Extensions will only be granted "after-the-fact" in extraordinary cases, e.g. involving an involuntary visit to the emergency room.

(c) I expect and strongly encourage(!) that you will work together on problem sets. Productive collaboration is an important part of science, and for most questions you should ask a peer before asking me. However, all work you turn in should be your own, reflecting your understanding of the problem, and all collaborators should be acknowledged. Copying work from any source will result in no credit for all involved. It is not acceptable to copy your peers' solution, to allow your solution to be copied from, or to work with your peer to prepare a single 'joint' solution set.

(d) You are encouraged to use whatever **static sources** you want to help you develop understanding and prepare your problem set. This means reading textbooks, class notes, scholarly papers, etc., whether in print or online. However, **responsive sources** such as posts on online fora or using chatGPT or other generative AI is disallowed unless explicitly allowed by me. Homework is not assigned simply to accumulate grades, but is the way you learn through practice – using AI to find solutions, like copying answers from others, undermines the learning process.

(e) Problem set solutions should show all necessary steps and be clearly explained and presented. Clear explanations require more than a bunch of algebra steps; use words and diagrams to explain your reasoning. For most students, this means that your turned-in **solutions are a cleaned-up and annotated re-write** of your scratch work.

- (f) Communication is key! If you need an accommodation due to sickness or a personal or family emergency, for a religious observance, or a disability, contact me as early as possible, and I will work with you to find a solution.
- (g) Come to office hours, it will be helpful!

Topics covered:

1. Ideal gases. Temperature, heat, and work. 1st law of thermodynamics. (2 lectures)
2. Probability (2 lectures)
3. Macroscopic irreversibility and multiplicity (2 lectures)
4. Entropy. 2nd and 3rd laws of thermodynamics. Phase space. (2 lectures)
5. Quasistatic processes. Heat engines and refrigerators. (2 lectures)
6. Boltzmann statistics and the canonical ensemble. The partition function. (3 lectures)
7. Helmholtz free energy and other Legendre transforms. (3 lectures)
8. Entropic forces and chemical potential. (2 lectures)
9. The grand canonical ensemble. (1 lecture)
10. Indistinguishable particles. Quantum statistical distributions. (3 lectures)
11. Fermi gases and Bose gases. (3 lectures)
12. Interacting systems. (2 lectures)

Days of note:

Wed 10/7: Midterm 1

Mon 10/12: Fall Break/Columbus Day/Indigenous People's Day

Wed 11/11: Midterm 2

Wed 11/25 & Fri 11/27: Thanksgiving break

Fri 12/11: Last day of classes

Wed 12/16: Review session 1:30-4:00 pm

Thu 12/17: Final exam 1:30-3:30 pm

Students with Disabilities: The University of Maryland is committed to providing appropriate accommodations for students with disabilities. Students with a documented disability should inform the instructor within the add/drop period if academic accommodations are needed. To obtain an Accommodation Letter prepared by Disability Support Service (DSS), a division of the University Counseling Center, please call 301.314.7682, e-mail dissup@umd.edu, or visit the Shoemaker Building for more information.

Medical excuses: Students are expected to inform the instructor in advance of medically necessary absences, and present a self-signed note documenting the date of the missed class(es) and testifying to the need for the absence. This note must include an acknowledgement that (a) the information provided is true and correct, and (b) that the student understands that providing false information to University officials is a violation of Part 9(h) of the Code of Student Conduct.

Religious Observances: It is the student's responsibility to notify the instructor within the first two weeks of class regarding any religious observance absence(s) for the entire semester.

Diversity: The University of Maryland values the diversity of its student body. Along with the University, I am committed to providing a classroom atmosphere that encourages the equitable participation of all students regardless of age, disability, ethnicity, gender, national origin, race, religion, or sexual orientation. Potential devaluation of students in the classroom that can occur by reference to demeaning stereotypes of any group and/or overlooking the contributions of a particular group to the topic under discussion is inappropriate.

Academic integrity: The UMD Honor Code prohibits students from cheating on exams, plagiarizing papers, submitting the same paper for credit in two courses without authorization, buying papers, submitting fraudulent documents and forging signatures. On every examination, paper or other academic exercise not exempted by the instructor, students must write by hand and sign the following pledge: **I pledge on my honor that I have not given or received any unauthorized assistance on this examination (or assignment).** Allegations of academic dishonesty will be reported directly to the Student Honor Council: <http://www.shc.umd.edu>

University policies: <http://www.ugst.umd.edu/courserelatedpolicies.html>